TUTORIAL 1 : INTRODUCTION TO OPERATING SYSTEM

Course	BCL1213 / BTL1213 / BIT1213 Operating Systems
Name	
Student ID	

QUESTIONS

1. Define an Operating System (OS) and explain why it can be considered a resource manager.

2. Explain the relationship between the following:

> Application → Operating System → Device Driver → Hardware

Using a printer or storage device as an example, describe what happens when an application requests access to the hardware.

3. An operating system provides users with three important capabilities:

> a) User interface
> b) Application execution
> c) Hardware resource allocation

Explain each capability and provide one practical example for each.

4. Two applications are running simultaneously:

> - Google Chrome requires memory and CPU time.
> - Microsoft Word requires memory and access to a storage device.

Explain how the OS acts as a resource manager in this situation.

5. Arrange the following developments in approximately chronological order:

- Linux
- UNIX
- Graphical User Interface
- Multiprogramming
- Mobile OS
- Early computers without OS

Then briefly explain how each development addressed changing computing requirements.

6. Why was multiprogramming an important development in operating-system history?

Explain the problem it attempted to address, rather than simply defining multiprogramming.

7. The slides identify portability and multitasking as important characteristics of UNIX.

Explain why portability would be valuable to an operating system.

8. A microkernel architecture keeps only major functions such as process and memory management in the kernel, while functions such as file management and I/O are placed in user space.

Why might an OS designer want to keep the kernel smaller? Explain your answer in terms of modularity, stability, or system organisation.

9. Based on the scenario, answer the following questions.

An airport is developing a new automated air-traffic monitoring system.

The system receives information from aircraft and sensors continuously. When critical information is received, the system must respond within a strict and predictable time requirement.

The development team is considering:

- Batch OS
- Time-sharing OS
- Network OS
- Real-time OS

Questions:

a. Which type of OS would be the most appropriate?

b. Should the system use a Hard RTOS or Soft RTOS? Explain.

c. Why would a batch operating system be unsuitable for this scenario?

d. Why is “fast processing” alone not sufficient to describe the requirement?

10. **“Modern operating systems have become more complex because computers have become more complex.”**

Do you agree with this statement?